

The Rock Cycle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Igneous, sedimentary, and metamorphic rocks change over time through processes such as cooling, weathering/erosion, compaction/cementation, heat/pressure, melting, and uplift.
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Round 1 - Retrieve It

1. How does magma or lava become igneous rock?

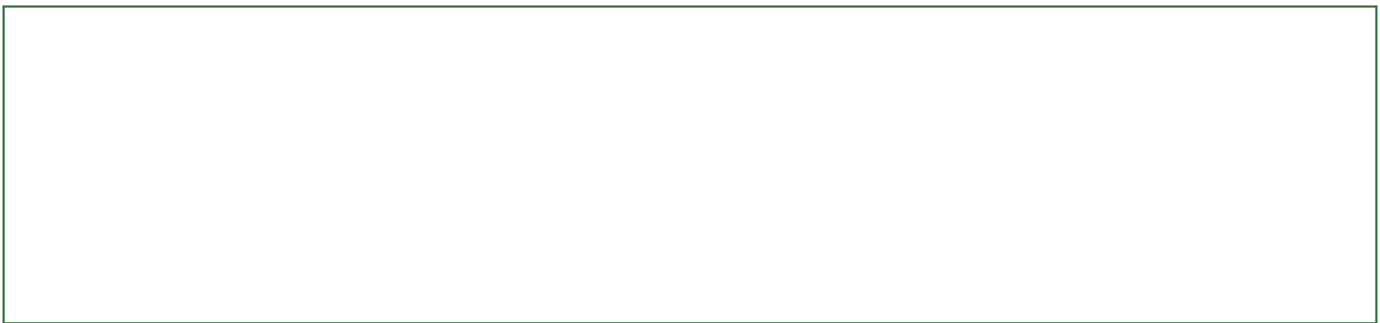
2. How do sediments become sedimentary rock?

3. How does existing rock become metamorphic rock?

4. What process can turn rock into magma?

Round 2 - Sketch It

Create a triangle with igneous, sedimentary, and metamorphic at the corners. Add arrows and process labels that show at least four ways rocks can change.



The Rock Cycle - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Granite melts underground. What material forms before a new rock type can form?

2. Sand is deposited, compacted, and cemented. What rock type forms?

3. Limestone is changed by heat and pressure without melting. What rock type results?

Round 4 - Reason from Evidence

Why is the rock cycle a cycle rather than a one-way sequence?

The Rock Cycle - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain The Rock Cycle in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect the rock cycle to plate tectonics. Which plate processes can expose, bury, melt, or deform rocks?

E**Extend**

Write one possible multi-step pathway that changes an igneous rock into a sedimentary rock and later into a metamorphic rock.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - The Rock Cycle

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. It cools and solidifies/crystallizes.
2. Compaction and cementation/lithification.
3. Heat and pressure change it without fully melting it.
4. Melting.

Sketch It - look for

Accurate processes such as cooling/crystallization, weathering/erosion/deposition, compaction/cementation, heat/pressure, melting, uplift.

Apply It

1. Magma.
2. Sedimentary.
3. Metamorphic.

Reason from Evidence - look for

Any rock type can follow multiple pathways and can change into several other types depending on processes and conditions; there is no single starting or ending point.

ACE - Extend look-for

Example: weathering/erosion -> sediments -> deposition/compaction/cementation -> sedimentary -> burial/heat/pressure -> metamorphic.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.